The Missing Organ: What Inference-Time Interventions Can and Cannot Buy in Open-Ended Generation

Roberto I. Ono Filho
Independent researcher
ono.roberto@gmail.com ORCID 0009-0006-8650-629X

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Abstract

A companion study [4] found that a base language model left to write for thousands of tokens gains judged surprise from one cheap operation, a new subject injected every few hundred tokens, and that the gains do not compose: no arm produced an integrated document, and on a problem with a verifier the same operation multiplied valid candidates without improving the best. Here we ask what would make local gains compose, by giving the loop, one at a time, the things a discovering mind has and the loop did not: a schematic memory instead of a verbatim one, a standing question instead of a neutral subject, selection over its candidates, a verifier inside its stream, and, finally, consolidation of its verified finds into its weights. All conditions are pre-registered and judged under the generated-only, fresh-only, document-level protocol of the companion study, or by a deterministic verifier. Three things worked and one did not. A schematic recap written by the model itself and re-injected with the new subject raises document-level integration above every earlier arm (+1.65 over the verbatim context, $p = 0.001$, ten premises; the highest integration score of the program), at no local cost. A standing open question extracted from the stream raises integration and development modestly ($p = 0.004$). Neither, nor their combination, moves development above about 2 on a 0–10 scale: no injection makes the whole advance. Selection over the interruption's candidates adds nothing to selection alone, and a verifier whose verdicts are injected into the notebook does not raise the ceiling either; instead the model writes its own verdicts, 16.4 fabricated "`# Verifier:`" lines per notebook against three real ones, a reward hallucination worth knowing about before putting a verifier in a prompt. Consolidation does something the others do not: three STaR/ReST cycles of LoRA fine-tuning on the model's own value-filtered candidates shift the distribution of what it writes on distributions it never trained on toward value (mean excess -1.3 to -1.5 points of excess over the base model on eight held-out variants, $p = 0.016$ at cycle 2 and 0.11 at cycle 3; -1.4 against a same-size random-consolidation control, $p = 0.035$), without collapsing diversity, and without moving the ceiling: no candidate beats the classic heuristics, and the best per variant is the same in every arm. The prior moves in mass, not at its frontier. We read the whole as a measurement of a boundary: at inference time the prior can be kicked (variation) and better remembered (integration), but it does not move; what moves it is what these loops lack, the updating of the model by what the world approved.

1 Introduction

Ono Filho [4] dismantled a cognitively inspired generation loop over 24 conditions and found that most of its effect on judged novelty lives in one operation, the *interruption*: a new subject injected every few hundred tokens into a stream whose literal repetition is damped. On windows of fresh generated text the interruption raises judged surprise by 1.2 to 1.4 points over habituation alone, on three base models, a post-trained one and two genres, and a pre-registered replication confirms it. The same study closed with three negatives. Read as whole documents, none of the streams builds anything (no arm above 2.5 on integration, development, coherence or surprise, 0–10, judged on 4,500 tokens with the injected text removed); a judge-gated review that decides when not to interrupt does not change that; and on online bin packing, where a deterministic verifier replaces the judge, the interruption multiplies the valid, distinct candidate heuristics a base model writes three- to fourfold without raising the quality of the best. The interruption is a variation operator, and "variation without selection is organized noise".

This paper takes up the question that study left: what would make local gains compose? We proceed by giving the loop, one piece at a time, the things a mind that discovers has and the loop did not, and measuring each under the protocol that the companion study showed to be necessary (generated-only windows, fresh-only estimates, a document-level reading, the premise as the unit of inference, a deterministic verifier where one exists). The pieces are (i) a *schematic* memory, in which the model carries a compressed recap of its own stream instead of the verbatim text that entrains it [1]; (ii) a *standing question*, in which the interruption points into the stream's own unresolved matter rather than away from it [7]; (iii) *selection*, a FunSearch-style evolution over the candidates the interruption multiplies [5]; (iv) a *verifier inside the stream*, whose verdict on each candidate is written back into the notebook; and (v) *consolidation*, a STaR/ReST cycle in which the model is fine-tuned (LoRA) on its own verified finds and generates again [2, 6, 8]. Every battery was pre-registered before it ran; the pre-registrations, amendments and results are in the dated laboratory notebook released with the code.

Contributions. (1) A schematic recap, written by the model and re-injected at each interruption, is the first intervention of the program to raise the document-level reading: integration +1.65 over the verbatim context and the highest integration score of any arm, at no cost on fresh windows. (2) A standing question raises integration and development modestly; its combination with the recap does not raise development further. Across all injections development stays near 2/10: we call this the progression wall. (3) Selection over the interruption's candidates and a verifier inside the stream both leave the ceiling where it was; the latter produces a new phenomenon, fabricated verifier verdicts, which we quantify. (4) Consolidation by LoRA on verified finds moves the candidate distribution on never-seen variants toward value (cycle 2 $p = 0.016$; vs a random consolidation control $p = 0.035$), keeps diversity, and leaves the ceiling where it was. (5) A reading of the whole: the bit distance between a fixed prior and a target is not shortened by inference-time stimulation; integration can be bought, progression cannot, and the organ that is missing is the one that updates the prior.

2 Setup

Generators and loop. Unless stated, the generator is Qwen3-30B-A3B-Base (mixture of experts, 8-bit, MLX); battery C uses Qwen3-8B-Base (8-bit) because it is fine-tuned. A cell is one premise (the ten premises of the companion study) continued for 4,500 tokens with the end-of-text token masked; habituation (windowed repetition penalty, window 512, factor 1.15) is on in every arm; interruptions fire on a clock every 300 tokens. Arms differ in what is injected and in whether the context is kept or rebuilt.

Judging. Windows of generated text only (96 tokens, starting 32 after any injected text, never crossing an injection), judged by Claude Opus 5 ($k = 5$, median) on surprise, connection and coherence with 600 tokens of context; windows flagged as self-copy (half their 12-token shingles seen earlier in the stream) are excluded from the fresh-only estimates. The whole stream,

arm (Qwen3-30B-A3B-Base, period 300)	integration	development	coherence	surprise
habituation, no interruption	1.40	1.10	1.50	1.40
verbatim context (clock300)	0.80	0.40	1.50	0.50
reset + new subject	2.00	1.60	2.20	2.50
schema: reset + recap + new subject	**2.70**	1.80	2.30	2.70
question: preserved + open question	1.50	1.10	1.90	1.00
agenda: reset + recap + open question	2.50	**2.00**	1.90	2.60

Table 1: Document-level reading (whole 4,500-token stream, injected text removed; Opus 5, $k = 3$; cell means over ten premises, 0–10). The schematic recap gives the highest integration of the program; no arm passes 2.0 on development.

injected text removed, is judged once per cell ($k = 3$) on integration, development, coherence and surprise. The premise is the unit: cell means, bootstrap intervals over cells, exact sign-flip permutation tests on paired cells, one-sided where pre-registered.

The problem with a verifier. Online bin packing in the FunSearch form: the model writes a notebook of `priority(item, remaining)` functions; the verifier packs held-out instances and reports the mean excess over the lower bound. Ten distribution variants (item ranges), training instances (seeds 100+v) and held-out instances (seeds 200+v). Battery C moves to a family of small-item distributions in which best fit is measurably beatable (Section 5).

3 Memory and tension: what the loop remembers and what it is asked

Arms. *Verbatim* (`clock300`): the context is kept across interruptions. *Reset* (`reset300`): the cache is rebuilt from the premise and the injected sentence only. *Schema* (`schema300`): at each interruption the model is asked, by a completion cue on its last 1,200 tokens, to write a two- to three-sentence past-tense recap; the cache is rebuilt from the premise, the recap ("*By then, this much had happened: . . .*") and the new subject. The recap counts as injected text (never inside a judged window; removed from the document). *Question* (`anomaly300`): the context is kept, and the injection is the open question the model itself extracts from its last 1,200 tokens ("*But one question remained: . . .*"). *Agenda* (`agenda300`): reset context, recap and question in the same injection. An anti-copy ladder retries the model's recap or question at higher temperatures when it reproduces the previous one (it tends to), and falls back to the plain rotation when it cannot; the oracle acts only after 200 tokens of stream.

Results. Table 1 gives the document-level reading. Pre-registered contrasts (one-sided, exact paired permutation over ten premises): M2, schema > verbatim on (integration+development)/2: +1.65 $[+1.15, +2.25]$, $p = 0.001$, positive on every dimension (integration +1.90, development +1.40, coherence +0.80, surprise +2.20; 10/10 premises on integration and surprise). M3, question > neutral subject: +0.70 $[+0.40, +1.05]$, $p = 0.004$, with integration and development each +0.70 ($p = 0.03$). M1, schema > reset on the composite: +0.45 $[-0.15, +1.05]$, $p = 0.13$, not supported; decomposed, integration +0.70 $[+0.10, +1.30]$ and development +0.20: the recap gives the model something to integrate and nothing to advance. M4, on fresh windows the recap costs nothing (surprise +0.15, connection −0.08, coherence −0.27, all n.s.; self-copy 0%; the schema arm's fresh windows, 3.90 / 3.22 / 6.60, are the best local scores of the program). The agenda arm (A1, primary: agenda > schema on development) gives +0.20 $[-0.40, +0.80]$, $p = 0.38$; the question changes the cast of each restart, not its progression (document judge: "restarts the premise over and over with new casts and genres").

Reading. Remembering the schema beats re-reading the text: the verbatim context, which entrains the model on thousands of tokens of its own past, reads worst as a whole; a reset reads better; a reset carrying a compressed memory reads best. But three injections (schema, question, agenda), all pre-registered, move integration and never development above about 2/10. We call this the progression wall and return to it in Section 6.

4 Selection and the verifier

Battery S: the interruption inside a selection loop. A FunSearch-style evolution (10 variants $\times$ 8 generations $\times$ 16 samples per generation; the prompt shows the current population; paired by variant) with and without the interruption's angle line injected into the prompt. S1 (gain of the champion on held-out instances): +0.0013 [-0.0015, +0.0047], $p = 0.38$. S2 (cumulative distinct valid candidates): 89.5 vs 95.1, $p = 0.86$ against the hypothesis. S3 (generation of first escape from best fit): +0.5, n.s. Under selection the interruption adds nothing, not even the diversity it added in a single stream: the population prompt already supplies what the interruption supplied.

Battery V: the verifier inside the stream. The notebook of the companion study's battery B, generated in chunks; whenever a `priority` function closes, the harness scores it on the training instances and writes the verdict into the notebook as a comment ("`# Verifier: priority_x scored mean excess 0.0812 on the training instances (worse than best fit, 0.0752); no improvement.`"). A second arm adds, every 300 tokens, a deterministic scoreboard ("`# So far:  N functions tried; the best is ...; best fit sits at ...`") and a value-anchored open question. Ten variants $\times$ two seeds; baselines are battery B's plain and angle arms re-scored in text order. V1 (held-out gain, feedback vs angle): +0.0003, $p = 0.48$. V2 (candidates improving on the notebook's own training best, the verifier's measure of progression): 0.1 vs 0.3 per notebook, opposite direction. V4 (scoreboard adds to feedback): nothing; the scoreboard arm writes fewer functions than every other arm (1.1 per notebook). Progression is near zero in every arm (0.1–0.3 improvements per notebook), with the standing caveat that best fit is near-optimal on these distributions.

Fabricated verdicts. The model wrote on average 16.4 "`# Verifier:`" lines per notebook in the feedback arm (6.2 with the scoreboard) against about three real ones: verdicts with invented scores and "new best of this notebook" that the harness never issued. Injecting the *format* of a reward signal into the context makes a base model generate its own reward; entrainment on the verdict is the code cousin of the self-copy the companion study found on narrative. Any design that places a verifier's output in a model's prompt should expect this and measure it.

5 Consolidation: moving the prior

Design (pre-registered 2026-08-21). Qwen3-8B-Base (8-bit), QLoRA (rank 8, 16 layers, lr 10^{-5}, batch 2, prompt masked, $\approx$4 epochs over the consolidation set). Domain: a family of small-item bin-packing distributions in which best fit is beatable by simple heuristics (headroom 0.3–0.5 points of excess; first fit ties or beats best fit), ten training variants and eight held-out variants that never enter any training set; verification on 20 $\times$ 200-item instances; a *find* beats the better of best fit and first fit by 0.001 on the variant's training instances. Cycle 0 is a shared base generation (3 notebooks $\times$ 1,200 tokens per variant, an angle comment every 250 tokens). Cycles 1–3: *attract* trains a fresh adapter on the finds plus the top-40 candidates by training excess from its own lineage; *random* trains on a same-size random sample of valid candidates

arm	cycle	candidates	mean excess	rel. to min(BF,FF)	best	distinct	copies
base	0	65	0.0633	+0.0322	0.0408	0.99	–
base	1	53	0.0562	+0.0251	0.0327	1.00	–
attract	1	235	0.0559	+0.0248	0.0319	0.98	10%
random	1	276	0.0708	+0.0397	0.0332	1.00	5%
base	2	42	0.0583	+0.0272	0.0317	0.97	–
attract	2	198	**0.0430**	+0.0119	0.0312	0.99	6%
random	2	274	0.0697	+0.0386	0.0310	0.99	8%
base	3	55	0.0644	+0.0333	0.0414	1.00	–
attract	3	214	**0.0516**	+0.0204	0.0311	0.99	7%
random	3	334	0.0653	+0.0342	0.0312	0.99	1%

Table 2: Battery C on the eight held-out variants (never in any training set; test instances): candidates per arm and cycle, mean test excess of valid candidates (cell means over variants), the same relative to the better of best fit and first fit, the best candidate, the distinct/valid ratio and the share of literal copies of the arm's training set. The value-filtered adapter lowers the mean without moving the best.

(the control that separates "trained on itself" from "trained on what is worth"); *base* generates without an adapter at the same seeds. Hypotheses, on the held-out variants at the final cycle: C1 (primary), attract < base on the mean test excess of valid candidates (the prior moved on variants it never saw); C2, attract < random; C3, higher find rate; C4 (two-sided), the diversity cost (distinct/valid, literal copies of the training set).

Results. Table 2 gives the held-out reading by arm and cycle. C1 (primary, final cycle): attract vs base on the mean test excess of valid candidates, -0.0129 [-0.0317, +0.0027], $p = 0.109$, not supported at $\alpha = 0.05$, with the direction consistent (7 of 8 variants better at cycle 3, 7 of 8 at cycle 2) and the cycle-2 contrast supported (-0.0153 [-0.0247, -0.0057], $p = 0.016$). C2: attract vs random, -0.0138 [-0.0252, -0.0024], $p = 0.035$, supported: the value filter matters. The random control hurts at cycle 1 (+1.5 points over base: consolidating mediocre functions makes the next ones worse) and is neutral by cycle 3. C3: strict finds are zero in every arm and cycle, and the best candidate per variant is the same in every arm (≈3.1% excess, the level of best fit): the ceiling did not move. C4: no collapse; the distinct/valid ratio is 0.97–1.00 throughout, literal copies of the training set 6–10% (attract); every adapter makes the model write 3.6–4× more functions per notebook. The adapter of the last cycle was taught mostly tight-bin functions (24 of 40 of the form `1/(r**2+eps)`, 13 best-fit-like): consolidation pulls the mass toward the best attractor the verifier has approved, not beyond it.

6 Discussion

What can be bought at inference. Across the two studies the ledger is now complete for the interventions we could build. Variation is cheap: the interruption raises fresh surprise by more than a point and multiplies valid candidates three- to fourfold. Integration can be bought with the right memory: a schematic recap carried across resets is the only intervention that raised the document-level reading, and it did so reliably. Progression cannot: three injections, selection over the candidates, a verifier in the stream and a judge-gated review all leave development near 2/10 and the verified ceiling where it was.

The bit distance. A generative model is a distribution, not a store; the question is not whether an answer is "inside" it but how much probability mass sits near it, how many bits of search separate the prior from the target. Inference-time stimulation reorganizes the surface of a fixed prior: it buys variation and, with a schematic memory, integration, but it does not move the mass. What moves it is what these loops lack and battery C supplies: updating the model by what the world approved. Battery C shows the lever and its present reach: three cycles of forty examples move the mass of the candidate distribution on never-seen variants toward the verified good, and no further than the best thing the verifier had already approved. This is also why the known successes of verified search [3, 5] pair a proposal distribution with a hard evaluator and many samples, not with a cleverer prompt.

Design rules. For long open-ended generation: damp repetition; interrupt with something new each time or reset the context; carry a compressed recap rather than the text; do not expect the whole to advance from any of this. For search with a verifier: spend on sampling and selection, not on stimulation; keep the verifier's verdicts out of the prompt or measure the fabrication they induce; if the prior must move, train on the finds.

7 Limitations

Ten premises and ten variants per battery; 8–30B base models at 8 bits; the document judge is a single LLM family at $k = 3$ and its absolute levels are low for every arm, so "integration 2.7" is a relative statement; the bin-packing distributions of batteries S and V have little headroom, which bounds what selection and feedback could show there (battery C's family was chosen for headroom for that reason); the oracle that writes recaps and questions is the generator itself and sometimes repeats itself, which the anti-copy ladder mitigates but does not remove; battery C is three cycles of forty examples on an 8B model with a rank-8 adapter, its primary contrast reaches $p = 0.016$ at cycle 2 and 0.11 at the pre-registered final cycle, its base arm varies by seed, and the strict find criterion admits lucky candidates (training finds lose on test), which is why the primary measure is the mean and not the best. The negative results are resolution-bounded nulls, not proofs of absence.

8 Conclusion

Given memory, tension, selection and a verifier, the interrupted loop composes better and advances no further. A schematic memory is worth having; a standing question helps a little; selection and in-stream verification add nothing to what they replace, and the latter teaches the model to invent its own verdicts. Consolidating verified finds into the weights is the one intervention that moved what the model writes on problems it had not seen, and it moved it toward what it already knew to be good. The organ that is missing from inference-time creativity is the one that learns.

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